

14	D	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$ $\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3pV}{Nm}} = \sqrt{\frac{3kT}{m}}$ $c_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(273.15 + 100)}{18(1.66 \times 10^{-27})}} = 719.04 = 720 \text{ m s}^{-1}$
15	D	Using $Q = mc\Delta\theta$